

Perceptual Quality Evaluation of Hazy Natural Images

Palak Mahajan , Vinit Jakhethiya, Pawanesh Abrol, Parveen K. Lehana, Badri N. Subudhi, and Sharath C. Guntuku 

I. INTRODUCTION

Abstract—Haze is an intrusion element that disrupts color fidelity and contrast of outdoor natural images, affecting their perceptual quality. The differential characteristics of hazy images compared to other natural images restrict the generalization of existing image quality assessment (IQA) algorithms. At the same time, efficient IQA algorithms for predicting the perceptual quality of naturally hazed images have not been proposed in the literature due to lack of a relevant dataset. To address this, we build the IIT-JMU Hazy Image Dataset comprising of 1000 high-definition hazy natural images consisting of diverse categories such as landscape, forests, roads, seascapes, and cityscapes, along with their subjective quality ratings. We present an analysis of existing natural-scene-statistics-based IQA algorithms on hazy natural images. In this article, we propose a convolutional-neural-network-based quality assessment algorithm for hazy natural images along with an IQA metric called deep learning-based haze perceptual quality evaluator (DLHPQE). The proposed DLHPQE efficiently predicts the perceptual quality of hazy natural images without a reference. Our results demonstrate that the DLHPQE outperforms existing state-of-the-art no-reference IQAs in terms of several performance parameters such as Pearson linear correlation coefficient, Spearman rank-order correlation coefficient, Kendall's rank-order correlation coefficient, and root-mean-square error.

Index Terms—Convolutional neural network (CNN), deep learning, hazy images, natural scene statistics (NSS), no-reference image quality assessment (NR IQA).

INTELLIGENT surveillance systems (ISS) such as road traffic control and autonomous vehicle driving, where pedestrian and vehicle detection are prime objectives, suffer significantly due to bad weather conditions such as haze. Haze obscures images captured by these ISS altering perception by diverging color and intensity of the scene [1]. In remote sensing applications, scattered atmospheric particles, such as haze, fog, mist, aerosol, etc., degrade the quality of a scene significantly. The uncertainty in weather conditions has urged research in perceptual quality estimation and enhancement models for hazy natural images. Eventually, many dehazing algorithms (DHAs) are proposed in the literature; consequently, it is required to have an automatic algorithm that can compare these algorithms. Gauging these hazed/dehazed images' visual quality is purely based on perceptibility, as such a quality measure is a prerequisite for image analysis in various vision-based ISS. Though research in dehazing gained traction, yet image quality prediction has been relatively unexplored. So, with escalating research in the domain of perceptual quality assessment, hazy image visual quality estimation is still mislaid.

DHAs to enhance visibility have been widely explored by researchers, and many DHAs have been proposed for this purpose. For instance, dark channel prior (DCP) proposed by He *et al.* [2] asserted that outdoor haze-free image is comprised of local minimum (close to zero) of the dark channel in at least one color channel. The dark channel has become the basis for many DHAs. Cai *et al.* [3] proposed a deep learning model, DehazeNet, to generate haze-free images by extracting haze features followed by quality improvement using a bipolar rectified linear unit (ReLU) activation function. Dudhane and Murala [4] applied a cardinal color channel along with a multichannel multiscale convolutional neural network (CNN) to suppress color distortion during dehazing.

The peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM) are commonly used quality metrics, but, in most cases, are incompatible with human perception [5]–[7]. Furthermore, the PSNR and the SSIM require complete information of the reference image, which are not available in practical scenarios. To effectively consider the visual perspective, image quality assessment (IQA) is used to estimate the image quality by propelling the human visual system (HVS). Objective IQA algorithms are categorized into three classes: full reference (FR), reduced reference (RR), and no reference (NR). The FR IQAs

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estimate quality based on the deviation between the original and distorted images. In RR IQAs, the distorted image's quality is estimated based on additional information given about the original image. NR IQAs estimate the quality of the distorted image without any prior knowledge of the original image.

Various IQAs for FR, RR, and NR have been proposed in the domain of natural, 3-D synthetic, screen content, and high-dynamic-range images. The SSIM [8], one of the earliest FR IQAs, computes the structural similarity between pristine and distorted images based on contrast, luminance, and structure.

For natural images, the natural scene statistics (NSS) model was made the primary basis for many IQAs such as NIQE [7], BRISQUE [6], and BIQI [9]. The foundation of these IQAs lies in modeling the statistical drift of features between natural and distorted images. Zhai *et al.* [10] presented an RR free-energy principle IQA metric based on brain theory to compute the quality of distorted images. Subsequently, HVS concepts were added to the free-energy principle to quantify the loss of naturalness yielding NFERM [11] blind quality metric.

Jakhetiya *et al.* [12] proposed an algorithm based upon the outlier detection for 3-D synthesized images. In [13], the authors developed an NR algorithm based on kernel ridge regression to estimate the perceptual quality of 3-D synthesized images along with image enhancement.

For hazy image quality assessment (HIQA), HDMHA [14] was designed to predict the density of synthetically incorporated haze in remote sensing images. Choi *et al.* [15] provided a prediction quality model FADE for measuring fog density by estimating the statistical irregularities between fogged and fog-free images. DHQI [16] estimated the quality of dehazed images generated from seven DHAs to demonstrate the potential of these DHAs against the image dehazing process. Also, Min *et al.* [17] predicted the quality of dehazed aerial and regular images, thereby providing an IQA for dehazing [3]. The BIQME algorithm [18], based on regression, extracts features such as contrast and brightness to measure visual quality providing enhancement of natural, dehazed, and low-contrast images. While still being a nascent research domain, prior work on HIQA is based on mostly synthetically created databases. The characteristics of synthetically created images are very different from the natural haze images [16]. For example, haze highly depends upon the depth of objects, and while creating synthetic images, with depth addition, a thick white cast often gets embedded, making it look unnatural.

To close this research gap, we propose an NR deep-learning-based haze perceptual quality evaluator (DLHPQE) IQA metric to learn end-to-end haze-related features from hazy natural images and estimate their perceptual quality. Also, the DLHPQE is designed to evaluate the perceptual quality of dehazed images, which can be iteratively used to discern haze in an image in the course of dehazing progressively. To facilitate this algorithm's evaluation, we develop a novel dataset of hazy natural images with subjective scores.

The DLHPQE is provided with ground-truth human opinion subjective scores collected from subjective testing, while the other parameters are learned layer by layer implicitly during the training of hazed images. The subjective score is added as

a parameter to provide human visual sensitivity. The DenseNet architecture [19] is used for feature learning and training over DLHPQE. The inclination behind adopting DenseNet over other CNNs is on compact feature maps with the maximal flow of information. The Adam optimizer is used for model tuning with DenseNet201. A linear regression layer replaces the Softmax layer to estimate quality predictions with mean square error (MSE) as the loss function. Since our approach is to study the natural aspects of haze, we have constructed a natural hazed image dataset IIT-JMU Hazy with 1000 images. With natural hazed images, the reference image might not be available all the time. Apparently, the infeasibility of reference haze images leads to NR haze IQA for measuring natural hazy images' quality in correlation with perceived visual quality. The motivation behind creating this dataset over the existing one has been explained in the later section. This premise is then compared with existing NR IQAs showing dominating performance over the IIT-JMU Hazy dataset.

The contributions of this article are summarized as follows.

- 1) We create a new database called IIT-JMU hazy image dataset (HID) consisting of 1000 hazy natural images with subjective scores. This is the only hazy natural image dataset with human annotations on perceptual quality scores to the best of our knowledge.
- 2) We analyze existing IQA algorithms' performance on the proposed IIT-JMU HID dataset.
- 3) We propose a DLHPQE IQA that predicts the quality of hazy images (in the IIT-JMU HID dataset) by applying the CNN to learn haze-relevant features. The proposed DLHPQE algorithm shows significantly better performance over existing NR IQAs in terms of Pearson linear correlation coefficient (PLCC), Spearman rank-order correlation coefficient (SRCC), Kendall's rank-order correlation coefficient (KRCC), and root-mean-square error (RMSE).

The rest of this article is organized as follows. Section II describes the background details of haze along with the statistical model defining haze, followed by a description of the existing datasets and the necessity of a new dataset. Section III introduces the IIT-JMU HID. The proposed IQA DLHPQE is presented in Section IV. The results are discussed in Section V. Finally, Section VI concludes this article.

II. BACKGROUND

A. Optical Model of Haze

The optical model of haze known as the Koschmieder model described through atmospheric scattering, defined by Narasimhan and Nayar [20], is demonstrated in Fig. 1. It depends on the attributes of the light that passes through the hazy environment. Mathematically, the hazy image $I(x)$ can be expressed as follows:

$$I(x) = J(x)t(x) + A[1 - t(x)]. \quad (1)$$

The hazy image $I(x)$ is composed of two components: $J(x)t(x)$ representing direct attenuation of the scene radiance

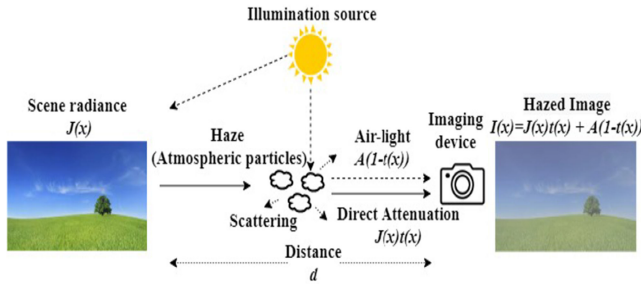


Fig. 1. Optical model of hazy image [3].

through transmission medium and $A[1 - t(x)]$ representing air-light, which is formed due to scattering effect. $J(x)$ is a haze-free scene image, with A being global atmospheric light, and $t(x)$ is the transmission medium in the atmosphere with values ranging from 0 to 1 along with x representing pixel intensity values. The transmission medium $t(x)$ can be represented by $t(x) = e^{-\beta d(x)}$, where β specifies the scattering coefficient in the transmission medium having $d(x)$ as the distance between the haze-free scene and the imaging object. The haze-relevant features, such as contrast, saturation, hue, and dark channel, characterize this atmospheric scattering. The basis of DHAs [3] is to identify these features for preprocessing followed by estimating the transmission medium $t(x)$ and the atmospheric light A .

B. CNN in Perceptual Quality

In recent years, deep-learning-based IQA models have emerged as promising techniques due to their strong representation potential. Diverse CNN models have been defined in literature for image perceptual quality evaluation. Zhang *et al.* [21] fine-tuned a bilinear model for blind IQA for synthetic as well as authentically distorted images. VGG-16 was applied along with stochastic gradient descent on the Waterloo Exploration Database. Bosse *et al.* [22] proposed a CNN-based FR and NR IQA for natural images based on VGGNet comprising ten convolution layers with additional two regression layers. Kim *et al.* [23] proposed CNN-based image quality measure, DIQA, consisting of two stages, where in stage 1, the model predicted the objective error map, followed by predicting subjective scores in the later stage, with the main objective being evaluating perceptual error maps. Wu *et al.* [24] proposed a blind IQA CNN based on hierarchical degradation observed in the HVS. The effect of distortions at different feature levels optimizes for quality prediction. Yan *et al.* [25] integrated NSS features in the deep learning model for quality prediction. A deep residual network is incorporated with the regression model for mapping between features and the quality score. Talebi and Milanfar [26] predicted the distribution for human opinion score by applying the CNN. Also, the proposed CNN-based model computed the aesthetic quality of images by applying the classification-based CNN. Zhang *et al.* [27] presented a CNN-based screen content image IQA by applying fuzzy classification. Visual quality assessment for both natural and synthetic regions was also estimated.

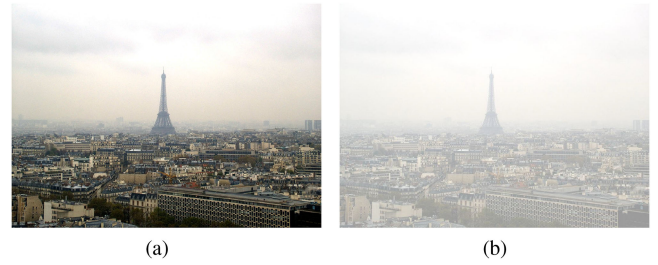


Fig. 2. Perceptual difference between naturally and synthetically hazed images. (a) Natural haze. (b) Synthetic haze.

C. Existing Datasets

Several datasets have been proposed to create new DHAs and also for HIQA. Table I encapsulates the particulars of the existing datasets. Ancuti *et al.* [28] created a synthetic dataset, D-HAZY, with induced homogeneous haze. FRIDA [29] and FRIDA2 synthetic databases with references are comprised of simulated urban road scenes. The O-haze dataset [30] contains hazy outdoor scenes created using different colors and reflectivity with an objective to evaluate the performance of various DHAs.

The LIVE defogging dataset [15] comprises 500 natural fogged and fog-free images, each with 100 available subjective scores for evaluating perceptual fog density. The Dense-Haze dataset [31], an extension of [30], consists of outdoor scenes of hazy and coinciding haze-free images. However, small-sized datasets have limitations while training deep-learning-based algorithms. The RESIDE dataset [32] comprises hazy synthetic images and few hazy natural images of around 0.4 million images with haze under different illumination conditions. The HazeRD dataset [33] comprises outdoor scenes with different weather conditions simulated by variation in fog levels, ranging from light fog to dense fog. Min *et al.* [16] constructed DHQ and SHRQ [17] datasets to calculate the quality of dehazed images.

- 1) From Table I, it can be observed that most of the existing datasets (such as FRIDA [29], D-Hazy [28], HazeRD [33], O-Haze [30], Dense-Haze [31], and SHRQ [17]) have hazed images created synthetically, which does not explicitly represent the properties of the hazy natural images, as shown in Fig. 2.
- 2) Although the LIVE Defogging dataset [15] has natural images with subjective scores, the number of images in this dataset is significantly low to generalize any algorithm.
- 3) With the DHQ dataset [16] comprising dehazed images, the images lack haze-relevant information, as these are constructed by applying various DHAs on hazy images.

To summarize, most datasets introduced in prior work use hazy synthetic images that vary in atmospheric light and haze density. Synthetically created hazy images are known to not accurately model natural hazy images. To justify this argument, we study the mean subtracted contrast normalized (MSCN) coefficients of the natural and synthetically created hazy images.

In general, MSCN coefficients follow a Gaussian distribution, and with the inclusion of distortion, they deviate from the

TABLE I
DESCRIPTION OF EXISTING HAZING DATASETS

Dataset	Year	No. of Images	Image Nature	Subjective Scores	Intended for	Remarks
FRIDA [29]	2010	480	Synthetic	Not Available	Fog detection with scene enhancement	Foggy images were developed using computer simulation fostering unnaturalness in the image.
LIVE Defogging [15]	2015	1,000	Real	Available (100 images)	Fog Density Evaluation	Dataset with collection of foggy and fog-free natural images.
D-HAZY [28]	2016	1,400	Synthetic	Not Available	DHAs Evaluation	Artificially induced haze imparted a white cast in the image, thereby hindering image quality.
RESIDE [32]	2017	0.4 M	Synthetic + Real	Available (20 images)	DHAs Evaluation	Very few images with subjective score is available to train a model for IQA analysis.
HazeRD [33]	2017	15	Synthetic	Not Available	Image Dehazing	A small dataset comprising of light to densely fogged images.
O-Haze [30]	2018	45	Synthetic	Not Available	DHAs Evaluation	Induced dataset comprised of heavily hazed images, created using haze machine.
Dense-Haze [31]	2019	33	Synthetic	Not Available	Image Dehazing	Poor perceptual quality of images due to dense white cast of artificial haze.
DHQ [16]	2019	1,750	Real	Available	Dehazing IQA	Formed by applying DHAs on real hazy images.
SHRQ [17]	2019	600	Synthetic	Available	Dehazing IQA	The white light used to induce artificial haze hampers the image quality.

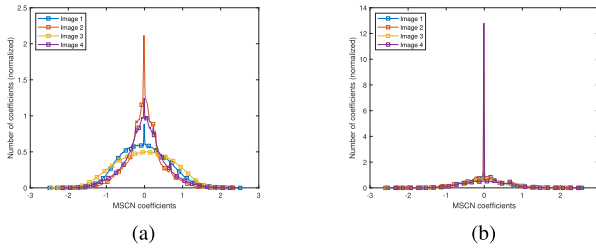


Fig. 3. Histogram of MSCN coefficients for IIT-JMU HID and FRIDA datasets. (a) Hazy natural image. (b) Synthetically hazed image.

Gaussian distribution. The MSCN coefficients for image $I_M(x, y)$ are defined as [13]

$$I_M(x, y) = \frac{I(x, y) - \mu(x, y)}{\sigma(x, y) + 1} \quad (2)$$

where $x \in 1, 2, 3, \dots, m$ and $y \in 1, 2, 3, \dots, n$ represent spatial indices with m and n as image dimensions. The local mean $\mu(x, y)$ and deviation $\sigma(x, y)$ are given as

$$\mu(x, y) = \sum_{k=-K}^K \sum_{l=-L}^L \omega_{k,l} I(x+k, y+l) \quad (3)$$

$$\sigma(x, y) = \sqrt{\sum_{k=-K}^K \sum_{l=-L}^L \omega_{k,l} [I(x+k, y+l) - \mu(x, y)]^2} \quad (4)$$

where ω denotes a 2-D Gaussian weighting window of size $K = L = 3$.

The histogram of MSCN coefficients for five images from our proposed dataset and the FRIDA dataset [29] with varying subjective scores is represented by Fig. 3. The histograms show that the inclusion of distortions (haze) does not alter the MSCN coefficients of synthetically created images significantly compared to the Gaussian distribution. For the proposed dataset, MSCN coefficients vary significantly with the increment of

haze. That is why, the MSCN-coefficient-based IQA algorithms cannot perform well on existing Haze IQA datasets.

From these arguments and literature survey, it is clear that only a few datasets are proposed in the literature for the purpose of IQA. Simultaneously, the existing datasets use synthetic images that do not replicate the characteristics of natural haze images. With this view, it is required to have a new dataset, which uses natural haze images and has subjective scores.

III. IIT-JMU HAZY IMAGE DATASET

The IIT-JMU HID comprises 1000 hazy natural scenes from copyright-free images collected from Flickr. The images are manually selected using keywords such as haze, sun-haze, aerial-haze, etc. All hazy images have high resolution with varied sizes ranging from 555×365 to 8057×5505 . Fig. 4 represents a few images from the IIT-JMU HID. The haze levels range from sparse to dense with diverse image scenes.

Furthermore, the IIT-JMU HID is a diverse dataset comprising varied views such as landscape, roads, forests, aerial view, etc. In order to justify that the proposed algorithm is able to handle images with different scene characteristics, we have distributed the dataset into various categories: Cityscape, Landscape, Skyscape, Seascape, and Miscellaneous. Fig. 5 represents the categorization of images over the IIT-JMU HID.

A. Subjective Testing Methodology

In order to evaluate the naturally hazy images, online subjective testing is conducted through Google forms. Online testing is similar to single-stimulus quality evaluation ITU-R BT.1082 standard, with the caveat that the latter does it in laboratory conditions. Participants were asked to rate the perceptual quality of naturally hazy images based on a five-grade quality scale (bad, poor, fair, good, and excellent) for each image. During the subjective testing, we have also asked the subjects about the level of haze for each image in the interval $[1, 5]$ to identify the relationship between the haze level in the image and the overall perceptual quality. A brief introduction to hazing was



Fig. 4. Example hazy images in the IIT-JMU Hazy Natural Images Dataset.

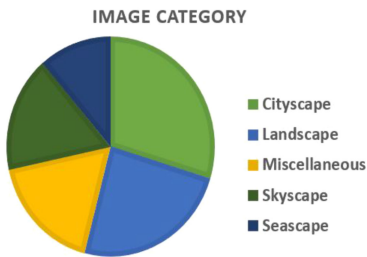


Fig. 5. Scenewise image categorization based on different characteristics.

TABLE II
SUBJECTIVE TESTING SETUP

Test Methodology	Description
Mode of testing	Online using Google form
Quality evaluation standard	Single-stimulus (SS)
No. of participants	14
No. of Sessions	10 with 100 image each
Session length	25-30 minutes

given to the participants before conducting subjective testing. Some hazy images (not a part of the subjective evaluation) were shown to explain the hazing phenomenon. The dataset images were divided randomly into ten subsets with 100 images each. The participants were requested to thoroughly observe an image at a viewing distance between 14' and 24' ITU standard. Fig. 6 shows a screenshot of the online subjective testing form.

The information about the device and its screen size and participant particulars such as expertise in image processing, gender, age, and education level were also asked during online subjective testing. Table II sums up the settings regarding subjective testing conducted.

B. Assessment of Subjective Score

Unprocessed subjective scores collected through the online evaluation are analyzed and used to obtain a mean opinion score (MOS) for every hazy image. The MOS for an image I can be

Fig. 6. Screenshot of the online subjective testing through Google form.

computed as

$$MOS_I = \frac{1}{N_J} \sum_{j=1}^J R_{IJ} \quad (5)$$

where R_{IJ} denotes the rating given by subject J to the image I , with N_J being the total number of ratings assigned to a particular image I .

In order to show that the proposed dataset is comprehensive, we have done the following three experiments.

- 1) For visualization of scores, we show the MOS over a single evaluation iteration containing 1000 images. Fig. 7 plots the MOS of 1000 hazy images along with the associated histogram. This figure shows that the proposed

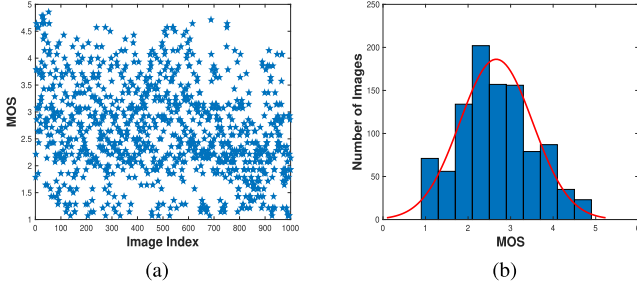


Fig. 7. (a) MOS distribution and (b) MOS histogram of 1000 images over the IIT-JMU HID.

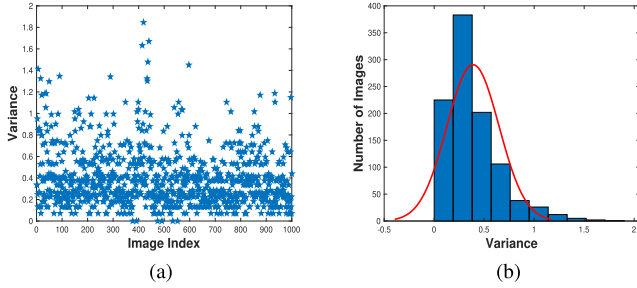


Fig. 8. (a) Variance distribution and (b) variance histogram of scores given by 14 subjects for each image of the IIT-JMU-HID.

IIT-JMU HID database has images with diverse perceptual quality (MOS). The quality scores span the entire range with dominating separation exhibiting perceptual quality.

- 2) During subjective testing, some of the subjects may be giving random scores to images. In order to verify that this is not the situation while creating the proposed IIT-JMU HID, we have calculated the variance of scores given by 14 subjects for each image shown by Fig. 8. In Fig. 8, we have calculated the variance of scores given by 14 subjects for each image. From this figure, it can be seen that the range of variance spans between 0 and 1.2 with values near zero for majority of images, which indicates slight diversions in-between subjective scores given by different subjects. Hence, the subjective testing results are quite legitimate for estimating haze perceptual quality analysis.
- 3) Also, haze density distribution is illustrated to depict coverage of varied density of haze. Fig. 9 represents the overall scatter of haze distribution over the entire range, demonstrating different haze levels in the dataset.

IV. PROPOSED ALGORITHM

This section describes the proposed referenceless quality assessment metric DLHPQE for hazy natural images on the IIT-JMU HID with an end-to-end learning CNN. The DLHPQE objectively evaluates the quality of the haze-degraded image. The proposed DLHPQE network architecture is depicted in Fig. 10. The basis of DLHPQE is to imbibe HVS-based sensitivity into the CNN using subjective scores. Initially, the

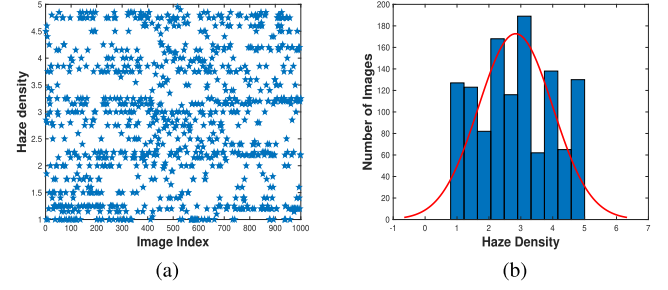


Fig. 9. (a) Haze density distribution and (b) haze density histogram of 1000 images of the IIT-JMU-HID.

subjective scores collected for perceptual quality have been mapped to a numeric range scale (1—Bad, 2—Poor, 3—Fair, 4—Good, and 5—Excellent). Influenced by the multilevel feature extraction with maximal information flow, DenseNet is used for the proposed DLHPQE framework. Better convergence with concatenation within layers of the dense block was the rationale for selecting DenseNet. Four dense blocks have been used to make the framework coherent, comprising of various layers at different blocks with transition layers in between. The predefined architecture of DenseNet201 has been used for weight training and model refinement. Also, for end-to-end learning, the subjective scores are fed directly along with corresponding images within the deep learning framework. The regression layer is attached at the end of the CNN to provide discrete values. These discrete values act as quality estimators for haze.

A. Architecture

For the DLHPQE network architecture, initially, DenseNet is fine-tuned on the IIT-JMU HID with 1000 images and perceptual quality subjective scores as input. DenseNet tends to provide connections between layers by allowing any layer to have connectivity with other subsequent layers. If we consider DenseNet comprising of layers l , with f being the nonlinear transformation applicable on each layer, then for each hazy image I , the input at any x th layer comprises preceding layer's feature maps expressed as

$$I_x = f_x[I_0, I_1, \dots, I_{x-1}]. \quad (6)$$

The term $[I_0, I_1, \dots, I_{x-1}]$ represents concatenation of feature maps from previous layers. The transformation functions f is generally compounded with three operations: batch normalization (BN), followed by a ReLU along with 3×3 convolution (Conv). Reshaping the size of feature maps i.e., downsampling, is done through splitting the architecture into multidensely connected dense blocks. The transition layers in-between these dense blocks perform convolution along with pooling operations. Pooling is performed by aggregating local quantified visual scores v_i for every image region i , giving a global quantified score V expressed as

$$V = \frac{1}{N_r} \sum_i^{N_r} v_i \quad (7)$$

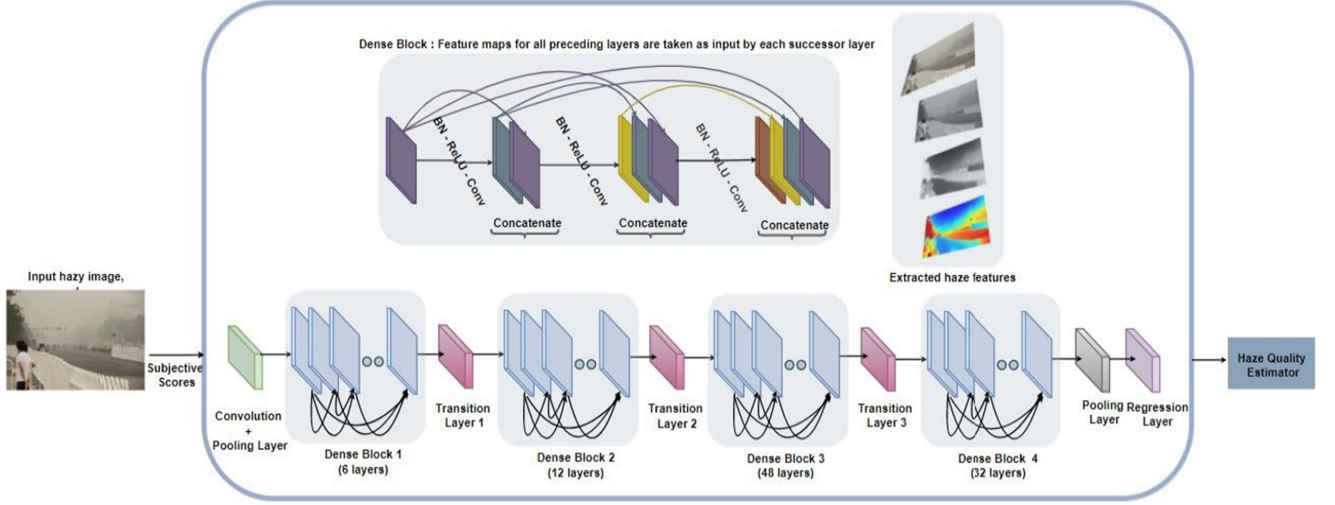


Fig. 10. Proposed DLHPQE architecture for the haze perceptual quality evaluator using DenseNet as a base CNN network. Each layer in dense block adds feature maps from previous layers. A sample example of a dense block is also encapsulated in the diagram for illustrating connectivity within inner layers while concatenating feature maps [19].

TABLE III
DLHPQE ARCHITECTURAL CONFIGURATION DETAILS [19]

Layers	Configuration	Output Size
Convolution	BN-ReLU-Conv 7×7 , stride 2	112×112
Pooling	max pool 3×3 , stride 2	56×56
Dense Block 1 (Dense Layers: 6)	BN-ReLU-Conv 1×1 BN-ReLU-Conv 3×3	56×56
Transition Layer 1	BN-ReLU-Conv 1×1 average pool 2×2 , stride 2	56×56 28×28
Dense Block 2 (Dense Layers: 12)	BN-ReLU-Conv 1×1 BN-ReLU-Conv 3×3	28×28
Transition Layer 2	BN-ReLU-Conv 1×1 average pool 2×2 , stride 2	28×28 14×14
Dense Block 3 (Dense Layers: 48)	BN-ReLU-Conv 1×1 BN-ReLU-Conv 3×3	14×14
Transition Layer 3	BN-ReLU-Conv 1×1 average pool 2×2 , stride 2	14×14 7×7
Dense Block 4 (Dense Layers: 32)	BN-ReLU-Conv 1×1 BN-ReLU-Conv 3×3	7×7
Regression Layer	Concat pool: Avg pool, Max pool BN-Linear-ReLU BN-Linear	1×1

where N_r represents the total number of regions sampled in every image I . The transition layers comprise a BN layer with 1×1 conv layer and a 2×2 average pooling layer. Also, a bottleneck layer of 1×1 conv layer is added before 3×3 conv to minimize the feature maps provided as input to the layers. The purpose of the transition layer and the bottleneck layer is to provide compactness in the network. The proposed DLHPQE architecture comprises four dense blocks along with three transition layers in between. The first dense block consists of six dense layers, followed by 12 dense layers in the second block with 48 and 32 dense layers in the third and fourth dense blocks defined by Table III. Each dense block comprehend with BN and ReLU accompanied with 1×1 and 3×3 conv layer defined in the following sequence: BN : ReLU : Conv(1×1) :

BN : ReLU : Conv(3×3). Accordingly, all dense layers are defined in this pattern. Likewise, BN and ReLU, followed by a conv and average pool, are defined for transition layers between the dense blocks. For positive and nonzero weights, the ReLU acts as a catalyst for weight activation with added stability factor ϵ .

For the final dense block, concatenated (concat) pooling with average pool and max pool is attached along with linear function for regression. The linear regression function is added in succession to truncate the output size to a single value. For the DLHPQE, the MSE has been defined as the loss function to be minimized during training as it is regression based. Let us consider x_i as input image with s_i as respective subjective score; the DLHPQE predicts the quality score $f(x_i, n_w)$, with n_w being the network weights. The loss function can be defined in terms of mappings between the predicted and ground-truth quality scores as

$$L = \frac{1}{n} \sum_{i=1}^n [f(x_i, n_w) - s_i]^2. \quad (8)$$

B. Transfer Learning

Transfer learning is employed on DenseNet201 for DLHPQE architecture training. The input to DLHPQE is an RGB image that is fed initially to the network. Since the dataset has been collected through online resources, data normalization is performed to transform input images of varied sizes to 256×256 . The subjective scores in the range [1,5] are forwarded to the DenseNet to learn the HVS sensitivity toward the haze. While training the network, each image is linked with the corresponding subjective quality score so that the network can process image features relatively to haze. The layers and the weights are adjusted according to DenseNet201 architecture while cutting down its classification layer, i.e., removing the softmax layer. Accordingly, two linear regression layers are added as the last

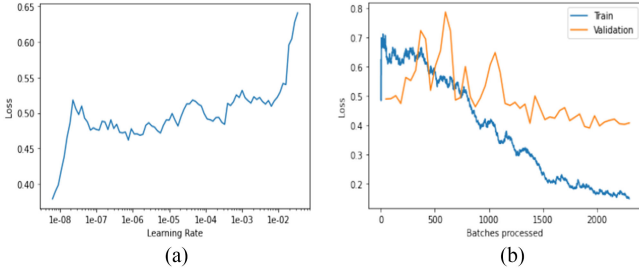


Fig. 11. Loss profile for (a) CLR and (b) training IIT-JMU HID on the DLHPQE during a k th cycle.

layer that provides a 1-D output, which gives the quality score prediction objectively. Since it is preferable to reduce the layers gradually, the inner nodes are reduced using the first regression layer with the successive application of the second regression layer, thereby providing a single prediction value at the end.

C. Training Method

The model is trained with the Adam optimizer by using gradient's variance with parameters $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\epsilon = 10^{-8}$ with weight decay of 10^{-2} . The dropout layer is added with a dropout rate of 0.25 and 0.5 for the first and second regression layers, respectively. The training involves k -fold dataset split, where $k = 4$; the hyperparameter, i.e., learning rate (LR), is setup using one cycle learning rate (CLR) scheduler to find out the LR with minimum loss. The CLR identifies steep loss areas and finds finer flatter minima; hence, the network converges faster. Generally, the LR selected is one order lower than the LR, at which the actual minimum loss occurs. The LR is set within the boundary $[1e^{-8}, 1e^{-2}]$ to find out the optimal LR at which the DLHPQE fits into minimum loss defined through cycle length less than training epochs. For the k th cycle, the loss rate for CLR learning and training DLHPQE is given in Fig. 11. Here, the DLHPQE gives minimum loss at $1e^{-6}$, as shown by Fig. 11(a), so we define $1e^{-7}$ LR to train the network. Fig. 11(b) plots training DLHPQE on the IIT-JMU HID with MSE defining the regression loss for aforementioned settings for batch size 16 with 50 epochs. Adam considers the sparse features, thereby giving a faster convergence rate [34]. While training the model, the spikes can be observed due to mini-batch gradient descent in Adam with a batch size of 16. With a total of 2000 batches processed, a minimum validation loss of 0.3903 occurs at the 41st epoch for this particular k th cycle. The trained model took an inference time of approximately 1.3647 s for predicting the quality score for a test image using Tesla GPU P100 16 GB.

V. EXPERIMENTAL RESULTS

A. Competing IQAs and Evaluation Criteria

To test the validity of the proposed IQA, we have compared the performance against 11 state-of-the-art existing NR IQAs over the IIT-JMU HID, since natural haze is an NR phenomenon as aforementioned. These state-of-the-art IQAs include BIQME [18], HDMHA [14], BRISQUE [6],

NIQE [7], PIQUE [35], BIQI [9], NFERM [11], OG-IQA [36], Fang15 [37], FADE [15], and SSEQ [38]. We have collated IQAs across various domains fabricated for discrete types of images such as natural images, screen content images, contrast-enhanced images, etc.

For performance evaluation, initially, the raw objective scores are converted according to the nonlinear logistic regression model comprising of five parameters defined according to ITU-R BT500 represented as

$$f(s_q) = \rho_1 \left(0.5 - \frac{1}{1 + e^{\rho_2(s_q - \rho_3)}} \right) + \rho_4 s_q + \rho_5 \quad (9)$$

where s_q and $f(s_q)$ are the predicted and the corresponding mapped quality scores and $\rho_i (i \in 1, 2, 3, 4, 5)$ represents the parameters observed for regression during curve fitting. The mapped values are then used to evaluate the performance using four IQA metrics: PLCC, SRCC, KRCC, and RMSE. PLCC is a prediction linearity measure, and RMSE measures prediction accuracy. SRCC and KRCC are nonparametric measures for the monotonicity of prediction. A good quality measure expects a value near 1 for SRCC, PLCC, and KLCC, while RMSE expects a value close to 0.

B. Performance Comparison

For evaluation purposes, the DLHPQE is trained on the IIT-JMU HID. The dataset is split randomly according to fourfold cross-validation. It is done to evaluate the unseen test image, which is not used for training purposes. Table IV represents the performance of the proposed DLHPQE IQA against the existing IQAs.

The DLHPQE is evaluated using fourfold cross-validation with set-out on mean values of PLCC, SRCC, KRCC, and RMSE. As anticipated from the table, the proposed DLHPQE achieves PLCC, SRCC, KRCC, and RMSE values as 0.6834, 0.6547, 0.4545, and 0.6282 surpassing the existing NR IQAs. From Table IV, we can infer the following aspects.

- 1) The traditional NR IQAs, defined for natural images based on the NSS model, are insensitive toward haze. FADE [15], however, gave a better quality prediction with PLCC, SRCC, KRCC, and RMSE of 0.3925, 0.3507, 0.2416, and 0.7877, respectively. With fog being one of the atmospheric phenomena like haze, FADE is competent with the proposed DLHPQE. The reason being FADE is designed to estimate perceptual fog. Hence, it can estimate the quality of hazy images to some extent.
- 2) HDMHA metric defined for haze estimation also fails with values 0.1661, 0.1286, and 0.0867 for PLCC, SRCC, and KRCC, respectively. Since it is defined to quantify synthetic hazing, it is inefficient to express the natural haze scenario. Hence, it cannot be applied to real-time haze detection scenarios as it is impervious to natural haze.

Predominantly, the proposed DLHPQE's prediction ability is at par to any NR IQA, giving higher values for coefficients PLCC, SRCC, and KRCC, with a comparatively lower value for RMSE. With the fact, the DLHPQE is completely referenceless,

TABLE IV
PERFORMANCE COMPARISON OF THE PROPOSED DLHPQE AGAINST 11 NR IQAs IN TERMS OF PLCC, SRCC, KRCC, AND RMSE ON THE IIT-JMU HID

No-Reference IQAs					
IQa	PLCC	SRCC	KRCC	RMSE	Oriented for
HDMHA [14]	0.1661	0.1286	0.0867	0.8446	Synthetic Hazed Images
PIQUE [35]	0.0942	0.0418	0.0291	0.8527	Natural Images
BIQI [9]	0.1809	0.1340	0.0913	0.8423	Natural Image
OG-IQA [36]	0.3549	0.3334	0.2314	0.8007	Natural images
NFERM [11]	0.3378	0.2662	0.1848	0.8061	Natural Images
Fang15 [37]	0.1888	0.1720	0.1165	0.8411	Contrast Distorted Images
BIQME [18]	0.2333	0.2207	0.1518	0.8328	Contrast Enhanced Images
BRISQUE [6]	0.1733	0.1583	0.1069	0.8436	Natural Images
SSEQ [38]	0.2460	0.2334	0.1603	0.8301	Distorted Images
NIQE [7]	0.2363	0.2392	0.1649	0.8322	Natural Images
FADE [15]	0.3925	0.3507	0.2416	0.7877	Fogged Images
Proposed DLHPQE	0.6834	0.6647	0.4848	0.6282	Natural Hazy Images

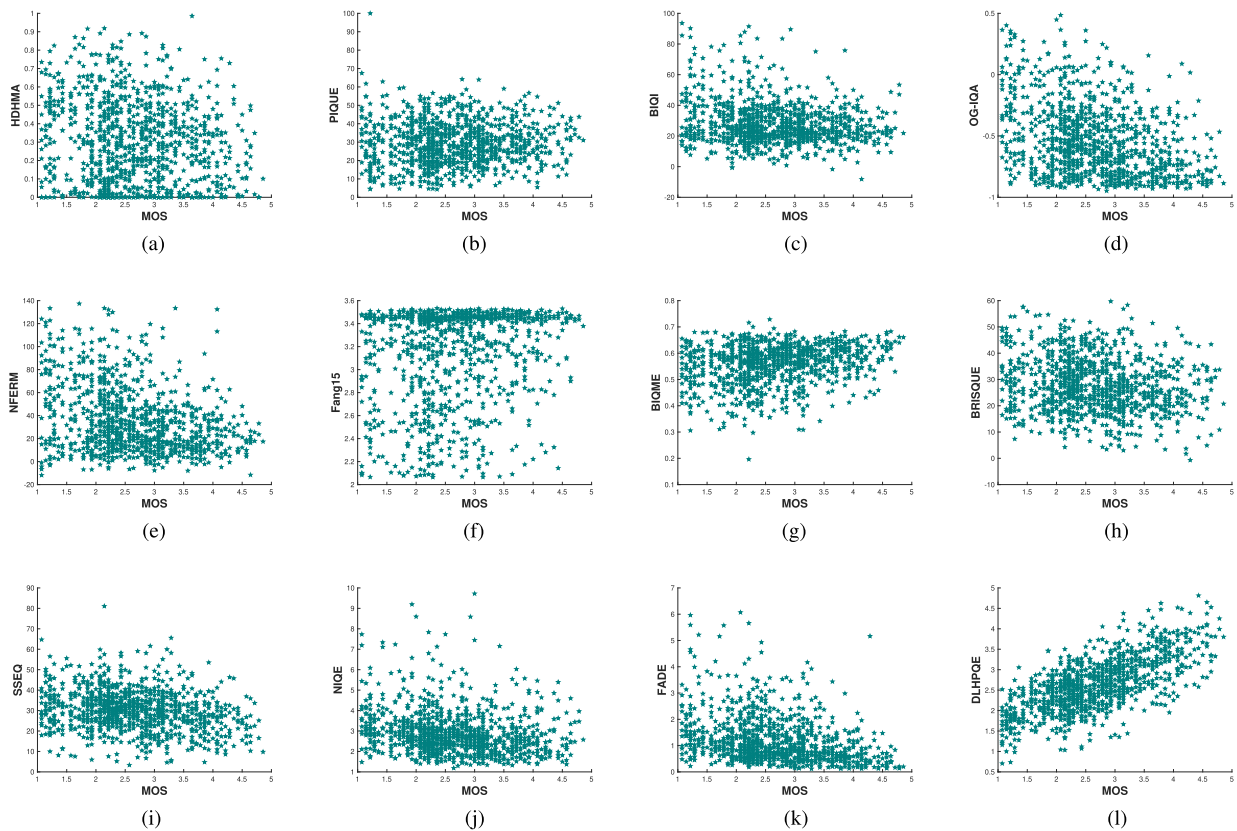


Fig. 12. Scatter plot for predicted objective scores of various IQAs against MOS values on the IIT-JMU HID. (a) HDHMA. (b) PIQUE. (c) BIQI. (d) OG-IQA. (e) NFERM. (f) Fang15. (g) BIQME. (h) BRISQUE. (i) SSEQ. (j) NIQE. (k) FADE. (l) DLHPQE.

since, for natural images, with-reference imaging is not feasible as aforementioned.

Also, DLHPQE's performance applied on the LIVE Defogging dataset [15] accord PLCC 0.6496, SRCC 0.6857 with KRCC 0.4631. This indicates that the proposed algorithm can be applied to measure the perceptual quality in foggy environments as well. Hence, the proposed algorithm is generic in nature and can be applied to other existing datasets as well.

Further, IIT-JMU HID is a diverse dataset comprising of varied views such as landscape, cityscape, skyscape, etc. In order

TABLE V
CORRELATION BETWEEN IMAGES WITH VARYING CHARACTERISTICS

IQa	No. of Images	PLCC	SRCC	KRCC	RMSE
Cityscape	297	0.702	0.688	0.501	0.620
Landscape	243	0.697	0.684	0.497	0.630
Miscellaneous	177	0.694	0.680	0.504	0.630
Skyscape	170	0.649	0.565	0.412	0.594
Seascape	113	0.639	0.609	0.448	0.583

TABLE VI
STATISTICAL SIGNIFICANCE COMPARISON OF THE PROPOSED DLHPQE WITH 11 EXISTING NR IQAS

IIT-JMU HID	HDMHA	PIQUE	BIQI	BIQME	Fang15	NFERM	SSEQ	NIQE	BRISQUE	FADE	OG-IQA
S	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1

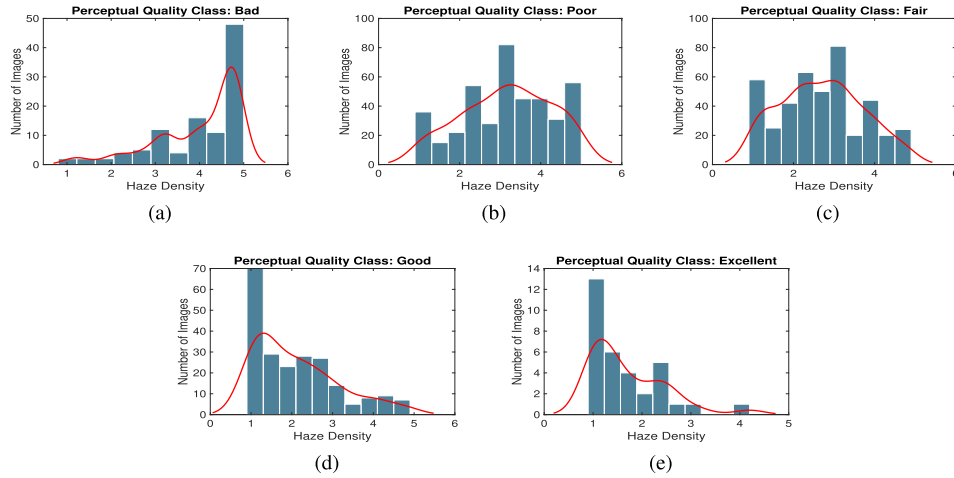


Fig. 13. MOS versus density profile for each MOS level. (a) Bad. (b) Poor. (c) Fair. (d) Good. (e) Excellent.

to justify that the proposed algorithm is able to handle images with different properties, we have shown results of the proposed algorithm on different kinds of images. **Table V** represents the distribution of dataset images into various categories and computation of IQA quality metrics. From this table, it is visible that the proposed algorithm is performing well for the images with different characteristics such as Cityscape, Landscape, Skyscape, Seascape, and Miscellaneous.

Fig. 12 exhibits the scatter plots for the predicted scores on the IIT-JMU HID manifested by various objective NR IQAs against the MOS of subjective scores for 1000 images. As outlined by the scatter plot, the data points in DLHPQE show linearity between predicted scores and MOS. Therefore, it is evident from the plots that compared with other NR IQAs, the DLHPQE shows more convergence. For the rest of the metrics, the plot of predicted quality scores over subjective scores is immensely dispersed, signifying their low performance. Hence, we can conclude that the predicted objective quality scores by the DLHPQE are congruous with subjective quality scores.

C. Statistical Significance

We also conducted an F-Test to testify the significance between the proposed DLHPQE algorithm with existing NR IQAs and assume that the predicted errors follow Gaussian distribution. The significance level considered is 0.05 with significance values +1 or -1 depicting the proposed DLHPQE better or worse than existing NR IQAs, while 0 significance value represents that the proposed DLHPQE is equally comparable to other NR IQAs. **Table VI** represents the observed values for the statistical significance test conducted between the proposed DLHPQE and other

TABLE VII
CORRELATION BETWEEN HAZE DENSITY AND PERCEPTUAL QUALITY

Metric	PLCC	SRCC	KRCC	RMSE
Value	0.4804	0.4626	0.3302	0.7665

NR IQAs. One can clearly remark from the table that the proposed DLHPQE is statistically better than all existing NR IQAs.

D. Relationship Between Haze Density and Perceptual Quality

During subjective testing, the subjective scores for haze density were also collected in the range [1, 5], where 1 represents mild haze, and accordingly, 5 signifies dense haze. Interestingly, the correlation coefficient between perceptual haze quality and haze density was only 0.4804 (as shown in **Table VII**). Their relationship can also be inspected graphically by plotting the haze density score for each MOS scale represented by **Fig. 13**. As represented by the plots, for bad-quality hazy images, an elevated curve toward the highest density score can be observed, whereas, for high-quality hazy images, the mildly dense haze was perceived as the peak of the curve procured at the lowest density scale. A stretched density curve followed them for average quality images. Hence, a PLCC of 0.4804 and curve plateau between haze density and haze quality implies that haze density does not directly affect perceptual haze quality. Simultaneously, a low level of haze does not significantly contribute to the perceptual quality estimation in hazy images.

VI. CONCLUSION

Estimating the quality of hazy natural images is an in-the-wild scenario with several real-world applications. We created the IIT-JMU HID comprising of 1000 hazy natural images accompanied by subjective testing. We proposed an NR quality estimator DLHPQE for haze based on deep learning. A CNN model was trained to learn haze-relevant features by providing human subjective scores to define haze sensitivity in HVS. The DLHPQE predicts the quality of the natural hazy scene. We validated the performance of the proposed DLHPQE against several state-of-the-art NR IQAs in hazy scenarios. Furthermore, the subjective scoring revealed that haze density is not directly proportional to haze quality. Since DCP is an essential feature in a haze, future work could include DCP features into the IQA framework to provide higher precision with increased sensitivity toward haze.

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